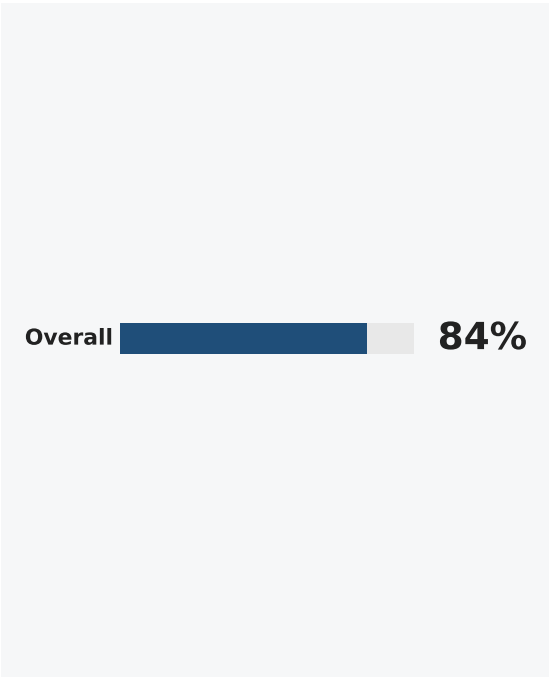
DroneVPS - Project Status

Marker-free GNSS-denied navigation for a 4.0 kg hexacopter. Target: 100-150 m AGL, 3-5 km out-and-back, no GNSS, no ground marker. Seventeen flights across 20-22 October. The VPS now holds position for minutes below 30 m and the map matcher works; above 34 m it does not hold, and that is the wall.



The matcher is twice as good as I had been reporting.
Scoring it needs the origin the node used. Derived once from the first row it wanders 21.36 m across a flight; per row it is exact. The same 27 fixes read p50 16.35 m one way and 8.58 m the other. The pilot said the pictures did not look like a 16 m error and was right.

The whole frame beats a centred square.
A square query is 0.95 of the SHORT side, so 44% of a 1280x800 frame never reached the matcher. On 40 frames at 85 m: p50 12.02 to 10.28 m, p90 30.43 to 23.86, for 10 points of fix rate. Widening the WINDOW was tried three ways and every one was worse.

Services survive a reboot now, and that was not true.
An ordering cycle had systemd deleting two of the three start jobs on every boot. Enabled, never attempted, no error anywhere. Found only because the aircraft was restarted and nothing came back.

Five changes have never flown.
CAM_C as the primary camera, the 6 m/s velocity cap, min_score 0.07, the IR laser and flood off, and the whole-frame query.

An item is 100% only when it has been verified in flight.

1. Platform and plumbing

Item		Status / evidence
Airframe, compute, sensors	**95%**	ZD850 hexa, CUAV X7 Pro, Jetson Orin, OAK-D Pro. The primary camera moved from CAM_B to CAM_C on 22 Oct and has NOT flown - every intrinsic, the distortion and the IMU rotation now read CAM_C.
Power and battery monitoring	**70%**	BATT_CURR_PIN was 4, which is the lidar's pin: the monitor was reading lidar volts x 24 and showing 16.25 A at idle. Now 17, and it reads 0.00 A - the Power A current line may simply not be wired. Voltage is correct at 24.8 V.
Vibration	**96%**	Not a problem anywhere. FC in flight VIBE p50 0.7/1.3/1.3, max 2.0, zero clipping against ArduPilot's 30. Camera 1.3x the FC. The 100 Hz IMU was folding 94% of its energy at Nyquist; now 400 requested, 150 delivered.
Attitude control and tuning	**85%**	ATC gains are ArduPilot stock on an 850 mm hexa. On GPS loiter the roll limit cycle is 0.9 Hz at 0.79 deg rms; on VPS the peak MOVES to 0.45 Hz at 2.41 deg and the demand itself grows, so the controller is reacting to our data. Autotune is the pilot's flight.
MAVLink routing	**94%**	One router owns the serial line, everything else is a UDP client. 0.71% of frames lose sync - low, unexplained. Listeners that stopped transmitting were being dropped after 30 s and going silently deaf; all of them heartbeat now.
Services and auto-start	**100%**	Proven by a real reboot: booted 09:07:18, all three started 09:07:37, zero restarts. An ordering cycle had been deleting two of the three start jobs on every boot - enabled, but never attempted.
Flight data recording	**97%**	66 columns per run plus saved frames at 4 Hz, fsynced every second. Frames only while armed. The recording is what found the camera accelerometer had never worked.
Preflight diagnostics	**92%**	One command before the props: preflight.py, and after_reboot_check.sh for the services. Both found real faults on their first run.

2. Position estimation

Item		Status / evidence
Optical flow	**95%**	The sideways sign in shift_to_ne was inverted and that was the runaway: velocity pointed up to 160 deg from truth. Fixed and confirmed in the air - angle 33/160/21 deg became 7/10/7.
Dead reckoning over distance	**85%**	flow_vel_frames 1 -> 40. Below 60 m the error at 0.5-2 m/s went 0.90-1.22 to 0.51-0.60 m/s, under the deadband, and that is what gave the 104 s hold at 16.5 m.
Gyro derotation	**95%**	Unchanged. Derotation runs off the FC attitude; the camera gyro is logged beside it for comparison.
Camera IMU configuration	**95%**	400 Hz requested, 150 delivered on the full pipeline. At 100 Hz the accel put 94% of its energy at 49.8 Hz, hard against Nyquist, and it was folding.
Mount alignment	**90%**	IMU-to-camera reads 179.6 deg off on BOTH sockets, so the accel mapping survived the CAM_C switch untouched. Not flown on CAM_C.
Map matching	**82%**	Working, and better than reported. Measured today on 40 frames at 85 m against the run's own receiver: p50 10.28 m, p90 23.86 m with the whole frame. Removing a constant offset leaves 8.58 m. Earlier figures near 16 m came from a reference that wandered 21 m.
Sensor fusion	**75%**	Map fixes are capped at 0.5 m each and arrive every 2 s, so the map can pull at most 0.25 m/s while drift at 80 m is faster.
Velocity to the FC	**60%**	Still the largest open fault. We were sending 25.47 m/s while the aircraft did 5.04; max_sent_mps 6.0 went in after the last flight and has NOT flown. EK3_SRC1_VELXY is 3, so the EKF fuses it.
Heading / yaw	**55%**	Unchanged. The compass in use is the M9N module's IST8310 and it is innocent of the glitch message - XKF4.SM p50 0.010 against a 1.2 gate.
Position injection as GPS2	**90%**	GPS_INPUT as GPS2 with velocity, AMSL from a real reference, and nothing of ours reaches the FC while the switch is on the receiver.

3. Flight controller integration

Item		Status / evidence
EKF origin and home	**100%**	On the VPS the origin comes from SCR_USER1/2 and never from the receiver, whatever fix it has.
Manual RC GPS1/GPS2 switch	**100%**	Channel 6, three positions, both directions announced on the radio. A deadlock that made the VPS unreachable from the transmitter was found and fixed: the switch now asks the node's liveness, not GPS2's fix.
Actually flying on the VPS	**70%**	Holds below 30 m - 167 s at 18 m, 104 s at 16.5 m, 74 s at 28.5 m, no failsafe. Above 34 m it drops to ALT_HOLD within a minute, every time.
Autonomous mission	**40%**	Not attempted on the VPS since the rebuild.
Mission speed ceiling	**70%**	LOIT_SPEED_MS 5, WP_SPD 4 - both reduced from the 4.7 defaults and confirmed still reduced.
Behaviour on RC loss	**100%**	Unchanged.

4. Range and coverage

Item		Status / evidence
Map coverage	**85%**	bhh.tif covers the take-off point with 1487 m to the east-west edge and 732 m north-south. 3.31 x 2.14 km at 0.509 / 0.437 m per pixel.
Sheet preparation and selection	**96%**	Unchanged. The east-west anisotropy correction is what makes XFeat match at all: 9 of 9 points with it, 0 of 9 without.
Altitude ceiling	**60%**	95.8 m reached and recorded. The target is 100-150 m, and the per-fix error improves with height - about 8 m at 96 m against 18 m at 83 m on the old reference.
3-5 km out-and-back	**15%**	Not attempted on the VPS. The whole of the gap.

5. Verification

Item		Status / evidence
Marker-free operation	**95%**	Unchanged.
Automated tests	**100%**	54 suites, all passing. Each names the real failure it pins.
Bench verification	**85%**	Offline replay against recorded frames is now the main tool - 88 recorded runs, 3194 frames above 60 m.
Flight verification	**90%**	17 flights across 20-22 Oct, logs 215-231, all backed up with their companion runs.
Offline replay before flight	**100%**	Recorded folders replay through the flying matcher.
Analysis tooling	**100%**	Unchanged.

6. Measured today

Measurement	Value	Where from
Map matcher, whole frame vs square	p50 12.02 -> 10.28 m, p90 30.43 -> 23.86 m	40 frames of vps_00084 at 85 m
Fix rate, that trade	52% -> 42% of frames accepted	the rate is what sets the system error
Widening the window	0.90 and 0.805 both WORSE; 0.805 p90 38.98 m	the query was the thing being cut, not the window
My own scoring error	p50 16.35 -> 8.58 m on the same 27 fixes	origin per row, not derived once; it wandered 21.36 m
Rotation was clipping the frame	100% kept at yaw 0, 81% at 30 deg, 62.5% at 90	only 327 of 901 frames sat within 15 deg
Three resamplings, not one	+60% fine detail recovered by one warp	resize, then rotate, then stretch - each one blurs
Camera is sharper than the map	9.9 cm/px against 50.9; high-freq 0.427 vs 0.312	but the map has nothing below 51 cm, so it cannot be used

6. Measured today (continued)

Measurement	Value	Where from
Matching above the map's GSD	51 cm 15% accepted, 40 cm 2%, 30 cm and finer 0%	upscaling the map makes blur, not detail
JPEG quality	90 to 35 moves inliers 12 to 11	the 5x downscale averages the artefacts away
Inlier tolerance 4.0 -> 6.0	fix rate 45% -> 80%, p90 24.39 -> 30.59 m	more inliers is not a better position
Residual constant offset	N +10.4 / E +8.8 m, 13.6 m long	removing it: p50 16.35 -> 9.60 m. Needs several flights
Commits since the last flight	4, and 78 since the SSD was lost	54 test suites, 66 CSV columns

7. Next

	What	Why
1	Fly it. Five changes have never been in the air.	CAM_C as primary, max_sent_mps 6.0 with its declared accuracy, min_score 0.07, the IR laser and flood off, and the whole-frame query. Two of those are large: the camera switch moved every intrinsic, and velocity is the channel that caused four EKF failsafes on 28 September.
2	Above 34 m is the wall.	Holds for minutes below 30 m and drops to ALT_HOLD within a minute above 34. Both ends are measured; what happens between them is not.
3	110 m, for the map.	The per-fix error falls with height and the ceiling so far is 95.8 m. This costs one flight and no code.
4	Settle the constant offset.	N +10.4 / E +8.8 m on one run. If it is the same vector on several flights it is the map's georeferencing and correctable; if it moves it is not. Only one run has enough fixes.
5	Battery current, under load.	0.00 A standing still. If the motors give real amps the wire is there; if not it is not, and every mAh figure is fiction until then.
6	The RGB camera, measured first.	2.45x the pixels on the same ground, and the map's 51 cm is the ceiling our query runs into. 1200p on the monos crashed the firmware, so bench it before assuming.

What gates the mission, lowest first: 3-5 km out-and-back 15%, autonomous mission 40%, heading/yaw 55%, velocity to the flight controller 60%, altitude ceiling 60%, actually flying on the VPS 70%, power monitoring 70%, mission speed ceiling 70%, sensor fusion 75%, map matching 82%. The aircraft holds position without GNSS for minutes below 30 m and the map is finding itself to 8-10 m. What is left is height, distance, and the velocity the flight controller is given.